MAT 324 HW08

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Problem 1. For a set X, collections $\mathcal{S}, \mathcal{S}_1, \mathcal{S}_2 \subseteq 2^X$, and $X' \subseteq X$, denote by $\sigma_X(\mathcal{S})$ the σ-algebra on X generated by $\mathcal{S}$ and define

$$\mathcal{S}|_{X'} = \{A \in \mathcal{S} : A \subseteq X'\}, \quad \mathcal{S} \cap X' = \{A \cap X' : A \in \mathcal{S}\}, \quad \mathcal{S}_1 + \mathcal{S}_2 = \{A_1 \cup A_2 : A_1 \in \mathcal{S}_1, A_2 \in \mathcal{S}_2\}.$$

Let X be any set.

(a) Suppose $\mathcal{S} \subseteq 2^X$ is a σ-algebra on X and $X' \in \mathcal{S}$. Show that $\mathcal{S}|_{X'} = \mathcal{S} \cap X'$ and that this collection is a σ-algebra on X'.

(b) Suppose $X_1, X_2 \subseteq X$ with $X = X_1 \cup X_2$ and $\mathcal{S}_1$ and $\mathcal{S}_2$ are σ-algebras on X_1 and X_2, respectively, such that $A_1 \cap X_2 \in \mathcal{S}_1$ for all $A_1 \in \mathcal{S}_1$ and $A_2 \cap X_1 \in \mathcal{S}_2$ for all $A_2 \in \mathcal{S}_2$. Show that $\mathcal{S}_1 + \mathcal{S}_2$ is a σ-algebra on X.

(c) Suppose $\mathcal{S} \subseteq 2^X$, $X_1, X_2 \in \sigma_X(\mathcal{S})$, and $X = X_1 \sqcup X_2$. Show that

$$\sigma_X(\mathcal{S}) = \sigma_{X_1}(\mathcal{S} \cap X_1) + \sigma_{X_2}(\mathcal{S} \cap X_2).$$

(d) Suppose $\mathcal{S} \subseteq 2^X$ and $X' \in \sigma_X(\mathcal{S})$. Show that $\sigma_{X'}(\mathcal{S} \cap X') = \sigma_X(\mathcal{S}) \cap X'$.

(e) Suppose $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ are measurable spaces, $X' \in \mathcal{S}$, and $Y' \in \mathcal{T}$. Show that $(\mathcal{S} \otimes \mathcal{T})|_{X' \times Y'} = (\mathcal{S}|_{X'}) \otimes (\mathcal{T}|_{Y'})$.

Solution. (a) If $A \in \mathcal{S}|_{X'}$, then $A \in \mathcal{S}$, and $A \cap X' = A \in \mathcal{S} \cap X'$, thus $\mathcal{S}|_{X'} \subseteq \mathcal{S} \cap X'$. On the other hand, if $A \in \mathcal{S} \cap X'$ then $A = B \cap X'$ for some $B \in \mathcal{S}$, hence $A \in \mathcal{S}$, and of course $A \subseteq X'$, so $A \in S|_{X'}$, implying the reverse inclusion $\mathcal{S} \cap X' \subseteq \mathcal{S}|_{X'}$.

(b) Since $\mathcal{S}_1$, $\mathcal{S}_2$ are σ-algebras, they contain $\varnothing$, so $\varnothing = \varnothing \cup \varnothing \in \mathcal{S}_1 + \mathcal{S}_2$.
If $A_1, A_2, \dots \in \mathcal{S}_1 + \mathcal{S}_2$, write $A_j = A_{j,1} \cup A_{j,2}$ where $A_{j,1} \in \mathcal{S}_1$ and $A_{j,2} \in \mathcal{S}_2$ for all $j \in \mathbb{N}$. Then

$$\bigcup_{j=1}^{\infty} A_j = \underbrace{\bigcup_{j=1}^{\infty} A_{j,1}}_{\in \mathcal{S}_1} \cup \underbrace{\bigcup_{j=1}^{\infty} A_{j,2}}_{\in \mathcal{S}_2} \in \mathcal{S}_1 + \mathcal{S}_2,$$

so $\mathcal{S}_1 + \mathcal{S}_2$ is closed under countable unions.
Lastly, let $A = A_1 \cup A_2 \in \mathcal{S}_1 + \mathcal{S}_2$ where $A_i \in \mathcal{S}_i$ for $i = 1, 2$. Since $A_2 \cap X_1 \in \mathcal{S}_1$ by assumption,

$$A_1' = A \cap X_1 = A_1 \cup (A_2 \cap X_1) \in \mathcal{S}_1.$$

Similarly, $A_2' = A \cap X_2 \in \mathcal{S}_2$. This implies

$$X \setminus A = (X_1 \setminus A) \cup (X_2 \setminus A) = \underbrace{(X_1 \setminus A_1')}_{\in \mathcal{S}_1} \cup \underbrace{(X_2 \setminus A_2')}_{\in \mathcal{S}_2} \in \mathcal{S}_1 + \mathcal{S}_2,$$

so $\mathcal{S}_1 + \mathcal{S}_2$ is closed under complements as well, making it a σ-algebra.

(c) Since X_1 and X_2 are disjoint, $\sigma_{X_1}(\mathcal{S} \cap X_1)$ and $\sigma_{X_2}(\mathcal{S} \cap X_2)$ satisfy the hypotheses of (c), so $\sigma_{X_1}(\mathcal{S} \cap X_1) + \sigma_{X_2}(\mathcal{S} \cap X_2)$ is a σ-algebra on X; it is contained in $\sigma_X(\mathcal{S})$ because the two σ-algebras consist of sets in $\sigma_X(\mathcal{S})$ and the summing procedure involves taking unions of sets that in this case are in $\sigma_X(\mathcal{S})$, producing more sets in $\sigma_X(\mathcal{S})$.

On the other hand, for each $A \in \mathcal{S}$, $A \cap X_1 \in \mathcal{S} \cap X_1 \subseteq \sigma_{X_1}(\mathcal{S} \cap X_1)$ and $A \cap X_2 \in \mathcal{S} \cap X_2 \subseteq \sigma_{X_2}(\mathcal{S} \cap X_2)$. Thus $(A \cap X_1) \cup (A \cap X_2) = A \in \sigma_{X_1}(\mathcal{S} \cap X_1) + \sigma_{X_2}(\mathcal{S} \cap X_2)$, so $\mathcal{S}$ is contained in the sum of the two σ-algebras, hence so is $\sigma_X(\mathcal{S})$, proving the reverse inclusion; thus $\sigma_X(\mathcal{S}) = \sigma_{X_1}(\mathcal{S} \cap X_1) + \sigma_{X_2}(\mathcal{S} \cap X_2)$.

(d) By part (c),

$$\sigma_X(\mathcal{S}) \cap X' = (\sigma_{X'}(\mathcal{S} \cap X') + \sigma_{X \setminus X'}(\mathcal{S} \cap (X \setminus X'))) \cap X'.$$

The only sets in the right-hand side that intersect X' are those in $\sigma_{X'}(\mathcal{S} \cap X')$ because the sets in the other addend are contained in $X \setminus X'$, which is disjoint from X'. Thus the right-hand side is exactly $\sigma_{X'}(\mathcal{S} \cap X')$, which is what we wanted to show.

(e) By (a) and (d),

$$\begin{aligned}(\mathcal{S} \otimes \mathcal{T})|_{X' \times Y'} &\overset{\text{(a)}}{=} (\mathcal{S} \otimes \mathcal{T}) \cap (X' \times Y') = \sigma_{X \times Y}(\mathcal{S} \times \mathcal{T}) \cap (X' \times Y') \overset{\text{(d)}}{=} \sigma_{X \times Y}((\mathcal{S} \times \mathcal{T}) \cap (X' \times Y')) \\ &= \sigma_{X \times Y}((\mathcal{S} \cap X') \times (\mathcal{T} \cap Y')) \overset{\text{(a)}}{=} \sigma_{X \times Y}((\mathcal{S}|_{X'}) \times (\mathcal{T}|_{Y'})). \qquad \square\end{aligned}$$

Problem 2 (Axler 5C #12)**.** (a) Prove that $\lim_{n\to\infty} \lambda_n(B_n) = 0$.

(b) Find the value of n that maximizes $\lambda_n(B_n)$.

Solution. By Axler's 5.44,

$$v_n := \lambda_n(B_n) = \begin{cases} \dfrac{\pi^{n/2}}{(n/2)!} & \text{if } n \text{ is even} \\ \dfrac{2^{(n+1)/2}\pi^{(n-1)/2}}{1 \cdot 3 \cdot \dots \cdot n} & \text{if } n \text{ is odd} \end{cases}$$

The subsequences v_{2n} and v_{2n-1} can be recursively defined as

$$v_2 = \pi, \quad v_{2n} = v_{2(n-1)} \cdot \frac{\pi}{n}, \qquad v_1 = 2, \quad v_{2n-1} = v_{2(n-1)-1} \cdot \frac{2\pi}{2n-1}.$$

(a) It suffices to show that v_{2n} and v_{2n-1} both converge to zero. If $n > 3$, then $n \geq 4 > \pi$, so

$$v_{2n} = v_6 \cdot \frac{\pi}{4} \cdot \dots \cdot \frac{\pi}{n-1} \cdot \frac{\pi}{n} < v_6 \cdot \frac{\pi}{n},$$

and the latter expression of course converges to zero. Similarly, for $n > 3$, $2n - 1 \geq 7 > 2\pi$, so

$$v_{2n-1} = v_5 \cdot \frac{2\pi}{7} \cdot \dots \cdot \frac{2\pi}{2n-3} \cdot \frac{2\pi}{2n-1} < v_5 \cdot \frac{2\pi}{2n-1} \to 0.$$

(b) By the inequalities in part (a), $\{v_{2n}\}_{n\geq 3}$ and $\{v_{2n-1}\}_{n \geq 3}$ are decreasing sequences, so the largest v_n is one of $v_1, \dots, v_6$. Furthermore

$$v_6 = \frac{\pi}{3} v_4 = \frac{\pi}{3} \cdot \frac{\pi}{2} v_2 = \frac{\pi^3}{6} \qquad v_5 = \frac{2\pi}{5} v_3 = \frac{2\pi}{5} \cdot \frac{2\pi}{3} v_1 = \frac{8\pi^2}{15},$$

so v_6 is the largest of the v_{2n} and v_5 is the largest of the v_{2n+1}; also, $v_5 > v_6$ because $\pi < 3.2 = 6 \cdot 8/15$, so

$$\max_{n \in \mathbb{N}} \lambda_n(B_n) = \boxed{\frac{8\pi^2}{15}}. \qquad \square$$

Problem 3. Let $m \in \mathbb{N}$, $(\mathcal{M}_m, \lambda_m)$ be the completion of the measure $(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_m, \lambda_1 \times \cdots \times \lambda_1)$ on $\mathbb{R}^m$, and $f \in \mathcal{L}^1(\lambda_m)$. For $x \in \mathbb{R}^m$ and $r \in \mathbb{R}^+$, denote by $B_r(x)$ the ball of radius r centered at x with respect to the standard metric on $\mathbb{R}^m$.

(a) Show that for every $\varepsilon > 0$ there exists a compactly supported continuous function $f_\varepsilon \colon \mathbb{R}^m \to \mathbb{R}$ such that $\|f - f_\varepsilon\|_1 < \varepsilon$.

(b) Define $f^* \colon \mathbb{R}^m \to [-\infty, \infty]$ by

$$f^*(x) = \sup_{r>0} \frac{1}{\lambda_m(B_r(x))} \int_{B_r(x)} |f| \, \mathrm{d}\lambda_m.$$

Show that f^* is Lebesgue measurable and that for all $c \in \mathbb{R}^+$,

$$\lambda_m(\{x \in \mathbb{R}^m \colon f^*(x) > c\}) \le \frac{3^m}{c} \|f\|_1$$

(c) Suppose f is continuous at $x_0 \in \mathbb{R}^m$. Show that

$$\lim_{r \to 0^+} \frac{1}{\lambda_m(B_r(x_0))} \int_{B_r(x_0)} |f - f(x_0)| \, \mathrm{d}\lambda_m = 0.$$

(d) Show that the equation in (c) holds for λ_m-almost every $x_0 \in \mathbb{R}^m$.

(e) Show that for λ_m-almost every $x_0 \in \mathbb{R}^m$,

$$\lim_{r \to 0^+} \frac{1}{\lambda_m(B_r(x_0))} \int_{B_r(x_0)} f \, \mathrm{d}\lambda_m = f(x_0).$$

Solution. (a) First, the analogue of Axler's 3.47 follows immediately from replacing $\mathbb{R}$ with $\mathbb{R}^m$ and "interval" with "product of intervals" throughout, as the preceding proposition 3.44 works in any measure space and we provided an outline for a proof of the analogue of Axler's 2.71 in class.

The generalization of Axler's 3.48, which is the statement of (a), follows in a similar manner using the generalized 3.47; the only detail that needs to be elaborated upon is the construction of a continuous function that is arbitarily close (in the L^1 norm sense) to $\mathbb{1}_B$ where $B = [a_1, b_1] \times \cdots \times [a_m, b_m]$ is a product of closed intervals. To do this, expand B to an open box $B' = (a_1', b_1') \times \cdots \times (a_m', b_m')$ such that $(a_k, b_k) \subseteq (a_k', b_k')$ and $\lambda_m(B' \setminus B) < \varepsilon$. Then, if $g_k \colon \mathbb{R} \to [0,1]$ is a continuous function that agrees with $\mathbb{1}_{[a_1,b_1]}$ on $[a_1, b_1]$ and has support contained in (a_1', b_1') (defined as in the picture in the proof of Axler's 3.48), the function

$$g \colon \mathbb{R} \to [0,1], \qquad g(x_1, \ldots, x_m) = g_1(x_1) \cdots g_m(x_m),$$

g is continuous, agrees with $\mathbb{1}_B$ on B, and $0 \le g - \mathbb{1}_B \le \mathbb{1}_{B' \setminus B}$, so

$$\|g - \mathbb{1}_B\|_1 = \int_{\mathbb{R}^m} g - \mathbb{1}_B \, \mathrm{d}\lambda_m \le \int_{\mathbb{R}^m} \mathbb{1}_{B' \setminus B} = \lambda_m(B' \setminus B) < \varepsilon$$

as desired.

(b) We show that $\{b \in \mathbb{R}^m : f^*(b) > c\}$ is open in $\mathbb{R}^m$ for every $c \in \mathbb{R}$, implying Lebesgue (in fact Borel) measurability; the proof is copied verbatim from Prof. Zinger's solution to 4A #9, with suitable changes for $\mathbb{R}^m$ instead of $\mathbb{R}$. We can assume $f \ge 0$. Suppose $b \in \mathbb{R}^m$ with $f^*(b) \in c$ and thus

$$\int_{B_r(b)} f(\,\mathrm{d}\lambda) > \lambda_m(B_{r+\varepsilon}(b)) \cdot c$$

for some $r, \varepsilon \in \mathbb{R}^+$. If $b' \in B_\varepsilon(b)$, then $B_r(b) \subseteq B_{r+\varepsilon}(b')$ and

$$f^*(b') \geq \frac{1}{\lambda_m(B_{r+\varepsilon}(b'))} \int_{B_{r+\varepsilon}(b')} f \, \mathrm{d}\lambda_m \geq \frac{1}{\lambda_m(B_{r+\varepsilon}(b'))} \int_{B_r(b)} f \, \mathrm{d}\lambda_m > c.$$

Recalling that we proved Axler's 4.4 for arbitrary metric spaces in class, the inequality is proven identically to Axler's 4.8, where 3^m replaces 3 in the numerator because $\lambda_m(3 * B) = 3^m \lambda_m(B)$ for any $B \in \mathcal{M}_m$.

(c) For any $r > 0$,

$$\frac{1}{\lambda_m(B_r(x_0))} \int_{B_r(x_0)} |f - f(x_0)| \, \mathrm{d}\lambda_m \leq \sup_{x \in B_r(x_0)} |f(x) - f(x_0)| \, ;$$

since f is continuous, the right-hand side gets arbitrarily small as r goes to zero, so the limit of the left-hand size as r goes to zero is 0.

(d) The proof is essentially that of Axler's 4.10, with Axler's 3.48 and 4.8 replaced by the results of (a) and (b) respectively, $\mathbb{R}$ replaced by $\mathbb{R}^m$, and open intervals replaced with open balls throughout. The only difference caused by these substitutions is that the 3 in equation 4.14 will be 3^m, but this is still a constant, so it makes no material change in the logic.

(e) This is a direct corollary of (d): for λ_m-almost every $x_0 \in \mathbb{R}^m$, we have the inequality

$$\begin{aligned}\left| \lim_{r \to 0^+} \frac{1}{\lambda_m(B_r(x_0))} \int_{B_r(x_0)} f \, \mathrm{d}\lambda_m - f(x_0) \right| &= \lim_{r \to 0^+} \frac{1}{\lambda_m(B_r(x_0))} \left| \int_{B_r(x_0)} f - f(x_0) \, \mathrm{d}\lambda_m \right| \\ &\leq \lim_{r \to 0^+} \frac{1}{\lambda_m(B_r(x_0))} \int_{B_r(x_0)} |f - f(x_0)| \, \mathrm{d}\lambda_m \overset{\text{(d)}}{=} 0,\end{aligned}$$

so the limit is equal to $f(x_0)$ for all x_0 at which (d) holds. □

Problem 4. Let $m \in \mathbb{N}$, $(\mathcal{M}_m, \lambda_m)$ be the completion of the measure $(\mathcal{M}_1 \otimes \cdots \otimes \mathcal{M}_m, \lambda_1 \times \cdots \times \lambda_1)$ on $\mathbb{R}^m$, and $A \in \mathcal{M}_m$.

(a) Show that

$$\lim_{r \to 0^+} \frac{\lambda_m(A \cap B_r(x))}{\lambda_m(B_r(x))}$$

equals 1 for λ_m-almost every $x \in A$ and 0 for λ_m-almost every $x \in \mathbb{R}^m \setminus A$.

(b) Suppose $m = 2$ and $\lambda_2(A) > 0$. Show that A contains three distinct points that form an equilateral triangle.

Solution. (a) Let $\delta_A(x)$ be the limit. We follow the proof of Axler's 4.24. The case where $\lambda_m(A) < \infty$ follows from #3(e) with $f = \mathbb{1}_A$. If $\lambda_m(A) = \infty$, then the argument from Axler's proof applies directly with $(-k, k)$ replaced by the ball $B_k(0)$ to use the finite-measure case.

(b) Suppose for the sake of contradiction that for some $A \in \mathcal{M}_2$ with $\lambda_2(A) > 0$ does not contain any three points forming an equilateral triangle. For any $p \in A$, let $f_p \colon \mathbb{R}^2 \to \mathbb{R}^2$ be the map sending $x \in \mathbb{R}^2$ to its rotation about p by $\pi/6$ (counterclockwise), so that $x \neq p$, p, and $f_p(x)$ form an equilateral triangle. Then, the only way that A does not contain three points forming an equilateral triangle is if A and $f_p(A)$ intersect only at p; this implies that, for any $r > 0$,

$$2\lambda_2(A \cap B_r(p)) = \lambda_2(A \cap B_r(p)) + \lambda_2(f_p(A) \cap B_r(p)) \leq \lambda_2(B_r(p)) \implies \lim_{r \to 0^+} \frac{\lambda_2(A \cap B_r(p))}{\lambda_2(B_r(p))} \leq \frac{1}{2}.$$

However, this contradicts the result of (a), which says that the limit above equals 1 for λ_2-almost every $p \in A$, hence cannot be at most $1/2$ for all $p \in A$. □